Q1) Class Hierarchy

```
class Person
        name, age
        input person()
              Read name
              Read age
End class
Class Student inherits Person
        reg No, marks[5], avg
        Calculate average()
            Sum = 0
            For i = 0 to 4
                Read marks[i]
                Sum = Sum + marks[i]
            avg = sum / 5
End class
Class Research Student inherits Student
        research Area
        display Complete Details()
            Print name, age
            Print regNo
            Print avg
            Print research Area
End class
main
      Create object rs of Research Student
            rs. input peson()
            Read rs. reg No
            rs. Calculate Average()
```

Read rs. research Area
rs. display Complete Details()
End main.

Q2 Complete the Missing:

```
BEGIN
READ N
READ ARRAY A
FOR i <- 0 TO N-1
    FOR j <- i+1 TO N-1
        IF A[I] = = A[J]
            A[J] = -1
        END IF
    END FOR
PRINT Modified ARRAY
END
```

Q3) Debug the Psuedocode

```
Class Area
METHORD area (length)
RETRUN length * Length
END METHOD
METHORD area(length, breadth
RETURN length * breadth
END METHOD
END CLASS
```

Q4) Trace output:

(i) Operation sum[value]

	Operation	sum[value]
1.	0-1	-1
2.	-1+2	1
3	1-3	-2
4	-2+4	2
5	2-5	-3

Final output = -3

Q5) Write Efficient PseudoCode:

```
Begin
READ N
READ ARRAY N
First ← -∞
Second ← -∞
FOR i<0 TO N-1
        IF A[I] > First
            Second ← First
            First ← A[i]
      ELSE IF A[i] > Second AND A[i] ≠ First
      Then
            Second ← A[i]
      END IF
   END FOR
Print Second
END
```

Q6) Convert problem statement into pseudocode.

```
Interface Payment method
        Pay(amount)
End Interface
Class Creditcard Implies payment
        method pay(amount)
    END method
END class.
CLASS UPI Implements payment
        Method pay(amount)
        END method
END Class
CLASS Net banking Implies payment
        method pay (amount)
    END Method
END Class

BEGIN
    READ choice amount
        IF choice =1
            P←Creditcard
        ELSE IF choice =2
            P←UPI
        END IF
            P.Pay(amount)
END
```

Q7) Fill in the blanks:

Class Dog inherits Animal

Animal obj ← new dog()

Q8) Correct Constructor.

```
CLASS Employee
DATA
    id
    name
Constructor (id, name)
        this.id ← id
        this.name ← name.
END Constructor
END Class.
```

(19) Case study (integrated problem).

```
Class vehicle
    Vehicle RegNo, Owner name
    METHOD display()
END Class
Class Car inherits Vehicle
    Fuel type
END CLASS
CLASS Truck inherits vehicle
    load capacity
END Class.
BEGIN
    READ N
    FOR i ← 1 TO N
        READ type
        If type = "car"
            obj ← New Car
        ELSE
            obj ← New Truck
        END IF
        Store obj
    END FOR
    FOR each obj
        obj.display()
    END FOR
END.
```